

PRD – AI Reel Generator from Video (MVP v1.0)

1. Product Overview

Product Name

AI Reel Generator

Purpose

Uploaded long-form videos se automatically high-quality reels generate karna, jisse manual editing ka time kam ho aur engaging short clips mil sakein.

2. Objectives

- Full video upload
 - Automatic reel generation
 - Smart highlight detection
 - AI-generated captions
 - Face-centered vertical crop
 - Thumbnail generation
 - Admin review before publishing
 - Continuous learning from reel performance
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3. Target Users

- Internal content team
 - Video editors
 - Marketing team
 - Content creators
 - Admin
-

4. User Flow

Upload Video

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Video Validation

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AI Processing Queue

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Speech to Text

↓

Scene Detection

↓

Highlight Detection

↓

Face Detection

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Vertical Crop (9:16)

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Caption Generation

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Thumbnail Generation

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Generate 10–20 Candidate Reels

↓

Admin Review

↓

Publish

5. Functional Requirements

Upload

Supported Formats

- MP4
- MOV
- MKV
- AVI

Maximum Size

Configurable

Video Processing

Generate

- 1080p
- 720p
- 480p

Extract

- Audio
 - Frames
 - Metadata
-

Scene Detection

Automatically detect

- Scene changes
 - Camera cuts
 - Motion changes
-

Speech Detection

Convert speech into transcript

Store

- Timestamp
 - Text
 - Language
-

Highlight Detection

AI should detect

- High emotion
 - Interesting speech
 - Strong expressions
 - Fast movement
 - Important keywords
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Reel Generation

Generate

- 15 sec
- 30 sec
- 45 sec
- 60 sec

Multiple candidates

Default

10–20 reels

Auto Crop

Aspect Ratio

9:16

Requirements

- Face always visible
- Speaker centered
- Smart camera movement

Subtitle Generation

Automatic subtitles

Languages

- English
- Hindi

Future

Multi-language

Thumbnail Generator

Generate

- 3 thumbnails

Admin selects one

6. AI Scoring Engine

Every candidate reel receives score

Example

Hook Score

Emotion Score

Speech Score

Motion Score

Face Visibility

Scene Quality

Replay Prediction

Overall Score

Only highest scoring reels suggested first.

7. Admin Review Panel

Display

- Video Player
- Candidate Reels
- Score
- Duration
- Thumbnail
- Transcript
- Suggested Title

Admin can

Approve

Reject

Trim

Regenerate

Publish

8. Reel Analytics

Store permanently

- Reel ID
- Source Video
- Category
- Tags
- Upload Date
- Publish Date
- Views
- Unique Views
- Watch Time
- Completion %
- Replay Count

- Skip %
 - Likes
 - Shares
 - Saves
 - Comments
 - Full Content CTR
 - Session Duration
 - Country
 - Device
 - Language
-

9. AI Learning Engine (Core USP)

AI continuously learns from user behaviour.

AI analyses

- Best categories
 - Worst categories
 - Best reel duration
 - Best upload timing
 - Best hooks
 - Best titles
 - Best thumbnails
 - Best keywords
 - Highest Full Content CTR
 - Highest completion rate
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10. AI Founder Dashboard

Every day AI generates insights

Examples

"Comedy category engagement increased 28%"

"25–35 second reels outperform 60-second reels."

"Videos uploaded between 7 PM and 10 PM perform best."

"Generate more content in Category X."

"Reduce uploads in Category Y."

11. Future Prediction Engine

AI predicts

- Next trending categories
- Recommended upload count per category
- Expected engagement score
- Estimated Full Content CTR
- Confidence score

Example

Recommended this week

1. Business Tips
2. AI Tutorials
3. Motivation

Expected CTR

34%

Confidence

91%

12. Database Tables

videos

video_processing_jobs

video_transcripts

video_scenes

video_faces

video_audio

reels

reel_scores

reel_transcripts

reel_tags

reel_thumbnails

reel_views

reel_events

reel_statistics

ai_predictions

ai_recommendations

admin_reviews

13. APIs

Upload Video

Get Processing Status

Get Candidate Reels

Approve Reel

Reject Reel

Publish Reel

Get Analytics

Get AI Suggestions

Get Trend Dashboard

14. Technology Stack

Backend

NestJS

Queue

Redis + BullMQ

Database

PostgreSQL

Storage

S3-compatible Object Storage

Video Processing

FFmpeg

Speech Recognition

Whisper

Scene Detection

PySceneDetect

Face Detection

MediaPipe

LLM

Llama / Qwen (pluggable)

15. Non-Functional Requirements

- Asynchronous processing
- Queue-based architecture
- Fault tolerance

- Retry failed jobs
 - Horizontal scalability
 - Processing logs
 - Monitoring dashboard
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16. Success Metrics

- Processing success rate > 99%
 - Average processing time < 10 minutes (30-minute video)
 - Admin approval rate > 60%
 - Reel completion rate
 - Full Content CTR
 - Watch time improvement over time
 - AI recommendation accuracy improvement month-over-month
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17. Long-Term Vision

Build a self-learning AI engine that not only generates reels but continuously improves based on user engagement. Over time, it should recommend:

- Which categories to create next
- Which reel style performs best
- Which titles, hooks and durations maximize watch time and full-content clicks
- Content strategy based on historical platform data